

On the Inductive Bias of Neural Tangent Kernels

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Problem

State-of-the-art neural networks are heavily over-parameterized. Optimization algorithms are crucial for learning models with good generalization properties.

Neural tangent kernel

In a certain overparameterized regime, the learning dynamics of gradient descent are governed by a certain kernel obtained at initialization.

The proposal of the paper

Studying the inductive bias of learning in such a regime by analyzing this kernel and the corresponding function space

Multiple works show that when a network is sufficiently over-parameterized, weights remain close to initialization during training. The model is then well approximated by its linearization around initialization:

$$f(x, \theta) \approx f(x, \theta_0) + \langle \theta - \theta_0, \nabla_{\theta} f(x, \theta_0) \rangle.$$

This regime where weights barely move has also been referred to as “**lazy training**” [Chizat et al.].

Note that weak differentiability (e.g., with ReLU activations) is sufficient when studying the limiting NTK.

L. Chizat, E. Oyallon, and F. Bach. On lazy training in differentiable programming. *NeurIPS*, 2019.

When the width of the network tends to infinity, the features of the linearized model tend to a limiting kernel K , called **neural tangent kernel**:

$$\langle \nabla_{\theta} f(x, \theta_0), \nabla_{\theta} f(x', \theta_0) \rangle \rightarrow K(x, x').$$

In this limit and under some assumptions, one can show that the weights move very slightly and the kernel remains fixed during training [Jacot et al.].

In case of finite number of model parameters convergence is allowed as long as the initial kernel matrix is non-degenerate.

A. Jacot, F. Gabriel, and C. Hongler. *Neural tangent kernel: Convergence and generalization in neural networks*, *NeurIPS*, 2018.

Consider a two layer network of the form:

$$f(x; \theta) = \sqrt{\frac{2}{m}} \sum_{j=1}^m v_j \sigma(w_j^\top x),$$

where $\sigma(u) = (u)_+ = \max(0, u)$ is the ReLU activation. The corresponding NTK is then given by:

$$K(x, x') = \|x\| \|x'\| \kappa \left(\frac{\langle x, x' \rangle}{\|x\| \|x'\|} \right),$$

where $\kappa(u) = u\kappa_0(u) + \kappa_1(u)$. κ_0 and κ_1 follow from standard calculations for arc-cosine kernels of degree 0 and 1.

Smoothness

- The kernel mapping for two-layer ReLU networks is not Lipschitz like κ_1
- It satisfies a weaker smoothness property similar to Hölder smoothness.

Approximation Properties

- The approximation properties are studied through the spectral decomposition of the NTK. The decay rate of eigenvalues indicates how well the RKHS can approximate different types of functions.
- NTK for two-layer ReLU networks has better approximation properties compared to other function classes.

Non-linear Operator M

This operator is applied point-wise to the feature maps and is essential for introducing non-linearity into the model. For two signals x and y , the operator M is given by:

$$M(x, y)[u] = \begin{pmatrix} \phi_0(x[u]) \otimes y[u] \\ \phi_1(x[u]) \end{pmatrix},$$

where ϕ_0 and ϕ_1 are kernel mappings associated with arc-cosine kernels of degrees 0 and 1, respectively.

NTK Feature Map for CNNs

$$\Phi(x)[u] = A_n M(x_n, y_n)[u],$$

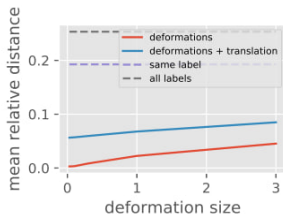
where A_n – pooling operator.

Stability to deformation

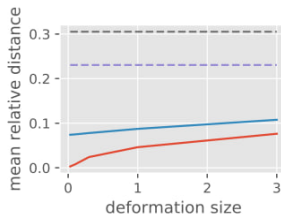
The NTK for CNNs provides a form of stability to deformations. The stability can be bounded, ensuring that small changes in the input lead to controlled changes in the output.

Limitations

Real-world CNNs with finite width may not fully exhibit the properties derived from the NTK in the infinite-width limit, leading to potential instabilities.



(a) CKN with arc-cosine 1 kernels



(b) NTK

- ① In the limit of a large number of neurons, NTK becomes constant.
- ② NTK determines how the network learns and allows analysis of function properties that can be efficiently learned.
- ③ The smoothness and approximation properties of NTK help explain why over-parameterized neural networks can generalize well, even though they have many parameters.
- ④ Less smooth behavior of NTK can be beneficial with large datasets.
- ⑤ Real networks with a finite number of neurons may not fully match NTK properties.
- ⑥ Important to balance stability and approximation when designing models.